

5 finishes in all the four rankings. Blender, Dropbox, BitBake, SageMath, YUM, OpenStack, Mercurial, Django, PyGTK, SciPy, etc, are some of the popular open source software developed with Python. Since Python is used so prolifically for open source project development, we can assign a high Openness Score to it.

Java

Java is a concurrent, object-oriented, general-purpose programming language. It is a popular choice for the development of enterprise, mobile and Web applications. OpenJDK (Open Java Development Kit) is a free and open source implementation of the Java Platform Standard Edition (Java SE), and since version 7 onwards it is the reference implementation of Java.

Java is an extremely popular programming language with the No. 1 rank in the TIOBE Index and PYPL Popularity rankings and a No. 2 rank in the IEEE Spectrum and RedMonk rankings. Some of the open source software developed with Java include Android, Weka, Apache Tomcat, Qt, Eclipse, NetBeans and Apache Hadoop. Operating systems are usually developed using C or Assembly language, but Java has made its presence felt there also. Though not very popular, the operating system JNode is fully developed using Java. So, for Java also we will assign a high Openness Score.

JavaScript

JavaScript is a high-level, dynamic, interpreted programming language. It is the programming language of the Web. It supports object-oriented, imperative and functional programming styles. The V8 JavaScript Engine is an open source engine developed by the Chromium Project for the Google Chrome Web browser. JavaScript is another programming language whose popularity is increasing in all the four rankings, and that makes it another recommended language to further your career. Mozilla Firefox, Mozilla Thunderbird, and MongoDB are some of the open source software developed with JavaScript. It has moderate importance in open source development and can be assigned a medium Openness Score.

Ruby

Ruby is a dynamic, general-purpose programming language. It supports multiple programming paradigms, including functional, object-oriented and imperative styles. Matz's Ruby Interpreter or Ruby MRI is the reference implementation of Ruby and uses the Ruby License, which is approved by the Free Software Foundation. The popularity of Ruby seems to be declining slightly over the years. What confuses me is the fact that though the rankings do not show any sinister decline in the popularity of Ruby, there are a lot of articles suggesting that it is dead

or going to die shortly – this is a phenomenon I couldn't comprehend, at least not with the data available. Popular open source projects developed using Ruby include Ruby on Rails, a server-side Web application framework and Merb, a precursor to Ruby on Rails. So, Ruby also has moderate open source relevance and can be assigned a medium Openness Score.

Swift

Swift is a general-purpose compiled programming language developed by Apple, mainly for the iOS and macOS platforms. It supports functional, object-oriented and imperative styles. The Integrated Development Environment for Swift is called Xcode, which is only



Figure 3: LLVM logo

a freeware using open source components. If it was the monkey that saved C#, for Swift the saviour is the dragon. Figure 3 shows the logo of LLVM, a very popular compiler infrastructure project; one of the Swift compilers is provided by LLVM. Since LLVM is open source software, Swift also qualifies as an open source programming language. As a side note, LLVM is an important open source project and the chief rival of GCC. If you find the time to explore LLVM, you will be rewarded. Calling Swift an open source programming language might be controversial but the language is gaining popularity in all the rankings and mastering it might be a wise decision. The Firefox iOS app, WordPress for iOS, etc, are some of the open source software written using Swift. The main problem with the open source projects developed with Swift is that most of them are built for the iOS or macOS platforms, which are tightly coupled with Apple products. Due to these reasons, we assign a low Openness Score to Swift.

So it's time to conclude the article and wait for angry protests! I am sure I have left out some of your favourite programming languages from this list. But do remember that you cannot remove any of the above-mentioned languages from the list due to various reasons. Of course, I could have added more languages to the list, but then it won't be a Top 10 list, then, would it? 

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